

PrithviNet

Mid-Evaluation Progress Report

Project Focus: Real-time National Environmental Monitoring System (CECB, Chhattisgarh)

1. Technological Stack

Frontend Ecosystem

- **React 19 & TypeScript:** Provides a highly responsive, strictly-typed user interface architecture avoiding runtime errors.
- **Vite:** Modern frontend tooling leveraging ES modules for rapid compilation and hot module replacement.
- **Tailwind CSS:** Custom, fully responsive, utility-first styling without large CSS overheads.
- **Leaflet (React-Leaflet) & Heatmaps:** Renders high-fidelity, interactive geographical heatmaps for active pollutant distribution points.

Backend Ecosystem

- **Python & FastAPI:** High-performance, highly concurrent ASGI web framework orchestrating complex REST, SSE, and WebSocket endpoints.
- **PostgreSQL + TimescaleDB:** Purpose-built time-series database architecture. Currently optimized to seamlessly handle massive query volumes (seeded with 104,000+ rows of granular 180-day hourly sensor history data).
- **SQLAlchemy & Asyncpg:** Fully asynchronous Object-Relational Mapping (ORM) handling smooth, scalable database connection pooling.
- **Redis:** In-memory key-value data store leveraged aggressively for caching high-load public map overviews, preventing database-level exhaustion during active polling.

AI & Predictive Forecasting

- **Facebook Prophet (Python):** End-to-end time-series forecasting algorithms that dynamically generate custom 3-day (48-72 hr) future predictions. Isolated binary `.pkl` models are independently trained natively for exact regional coordinates based on specific, unique node datasets.

2. Key Implemented Features

- **Authentic Geo-Spatial Mapping:** Digested officially verified Government datasets to accurately plot precise geospatial nodes (ex. Bhilai, Raipur, Tumidih) onto a dynamic layout rendering exact latitude/longitude clusters.
- **Strict Indian CPCB AQI Standardization:** Rebuilt core backend mathematical equations overriding broad global standards. Now structurally translates raw $\text{\$PM}_{2.5}$ ($\text{\$}\mu\text{g}/\text{m}^3$) metrics into the authorized Indian Air Quality Index. Actively triggers certified visual and semantic categories (Good, Satisfactory, Moderate, Poor, Very Poor,

Severe).

- **Real-Time Data Streaming:** Implemented persistent frontend background network pooling processes (10-second `setInterval` streams) intricately patched into asynchronous backend routers to keep visual interface nodes completely synchronized as native metrics trigger.
 - **Granular Historical Data Seeding:** Completely decoupled default structural databases to ingest natively formatted, structured `.csv` records, laying ground for high-accuracy ML simulations extending back 6 months.
-

3. Data Integrity & Deployment Structuring

Robust structural decisions include separating backend dependency injection bounds (via internal `.venv` instances), configuring active data-caching layers (Redis flushes triggered exactly on structural schema modifications), mapping full `docker-compose.yml` clusters, and providing independent build chains ensuring rapid progression towards production readiness.